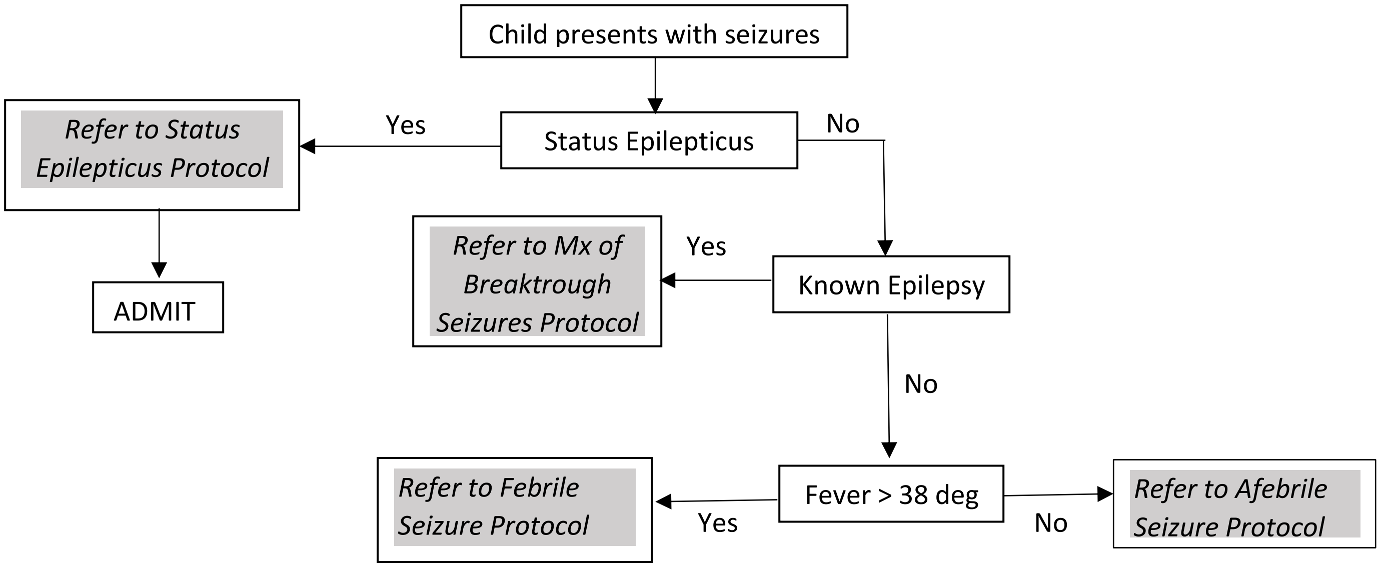
MANAGEMENT OF ACUTE SEIZURES IN CE



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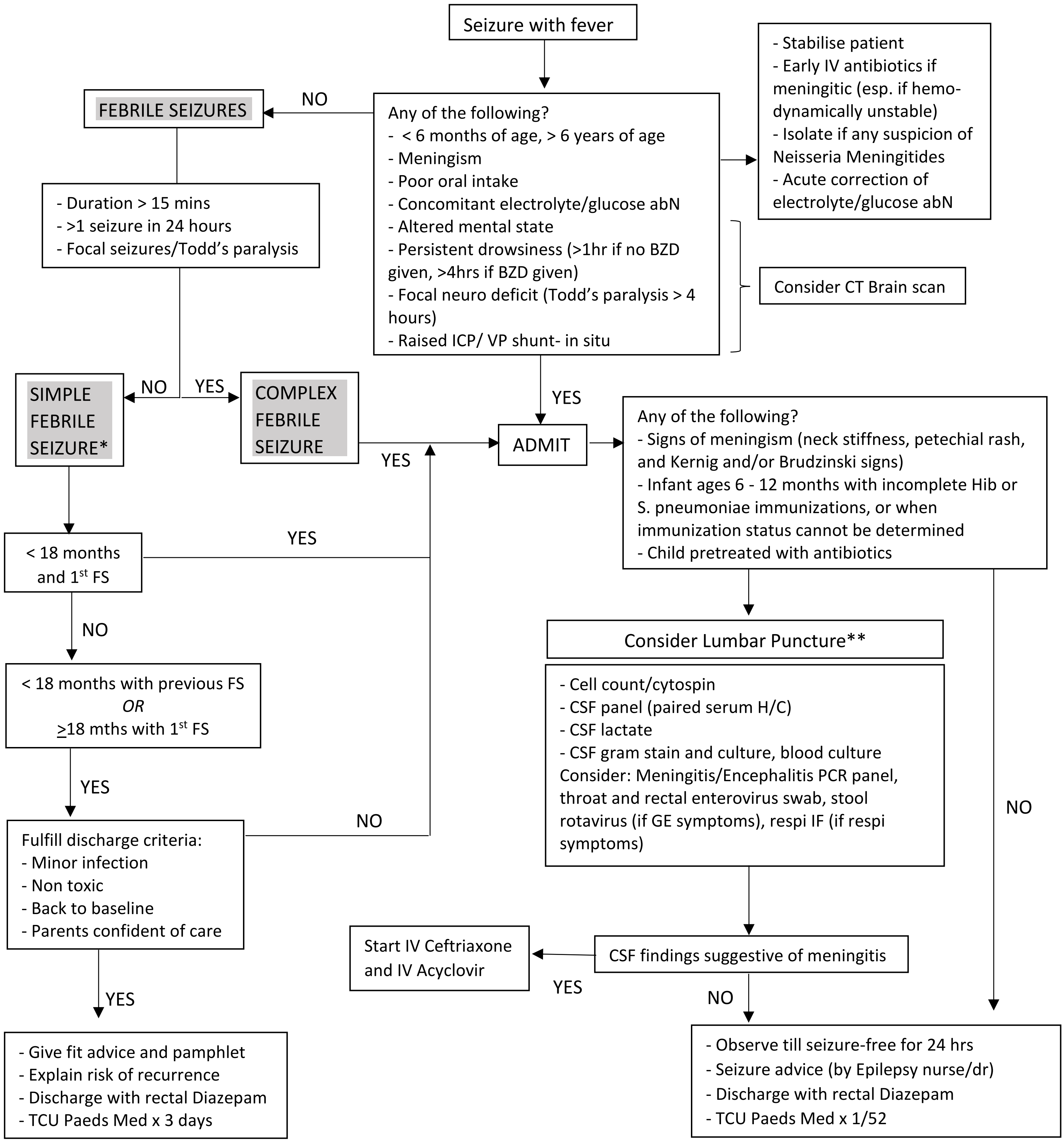
- approved by A/P Ong Hian Tat
February 2018

PAEDIATRIC FEBRILE SEIZURES/SEIZURES WITH FEVER PROTOCOL



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*No investigations are required for simple febrile seizures unless source of fever cannot be determined
**Discuss with consultant if there are any doubts about the need for LP and antibiotics

FEBRILE SEIZURES

Febrile seizures are the most common seizure disorder in childhood, affecting 2–5% of children. Febrile seizures are seizures that occur between the age of 6 and 60 months with a temperature of >38 °C, that are not the result of central nervous system infection or any metabolic imbalance, and that occur in the absence of a history of prior afebrile seizures.

Definitions

A Simple Febrile Seizure is a

- Generalized, usually tonic-clonic
- Attack associated with fever >38 °C
- Lasting not more than 15 minutes
- Single episode

A Complex Febrile Seizure is a seizure that has one of the following characteristics:

- Prolonged duration (more than 15 minutes)
- Focal seizure/Todd's paralysis
- Recurrent seizures within 24 hours

Febrile Status Epilepticus is a febrile seizure lasting more than 30 minutes.

Important Differential Diagnosis

- CNS infections e.g. meningitis or encephalitis
- Influenza associated encephalopathy of infancy and childhood (e.g. acute necrotizing encephalitis of childhood [ANEC]).This can also occur with other respiratory viruses, though much more rarely
- Drug intoxication
- Febrile rigors/delirium
- Syncope with reflex anoxic seizures during febrile illness (aggravated by dehydration etc)
- Epileptic seizures precipitated by the febrile illness in a patient with existing epilepsy
- Febrile Infection Related Epilepsy Syndrome (FIRES)

History and Physical Examination

A full history including developmental milestones, history of head trauma and a family history of seizures or developmental delay/regression is essential.

Important Clinical Signs to look out for include

• Vital signs

• Signs of meningism e.g. neck stiffness, photophobia, Kernig's sign, Brudzinski’s sign

• Signs of raised ICP e.g. altered level of consciousness, papilloedema, abducens nerve palsy, Cushing’s triad

• Neurocutaneous stigmata

• Rashes e.g. vesicles, petechia

Management aims:

1. Stop the convulsion

a. IV Lorazepam 0.05- 0.1mg/kg/dose over 2 mins

b. If no intravenous access:

- Rectal Diazepam 0.3-0.5mg/kg/dose
- Intra-muscular Midazolam 0.2mg/kg/dose

2. Assess if seizure was febrile seizure, exclude CNS infection

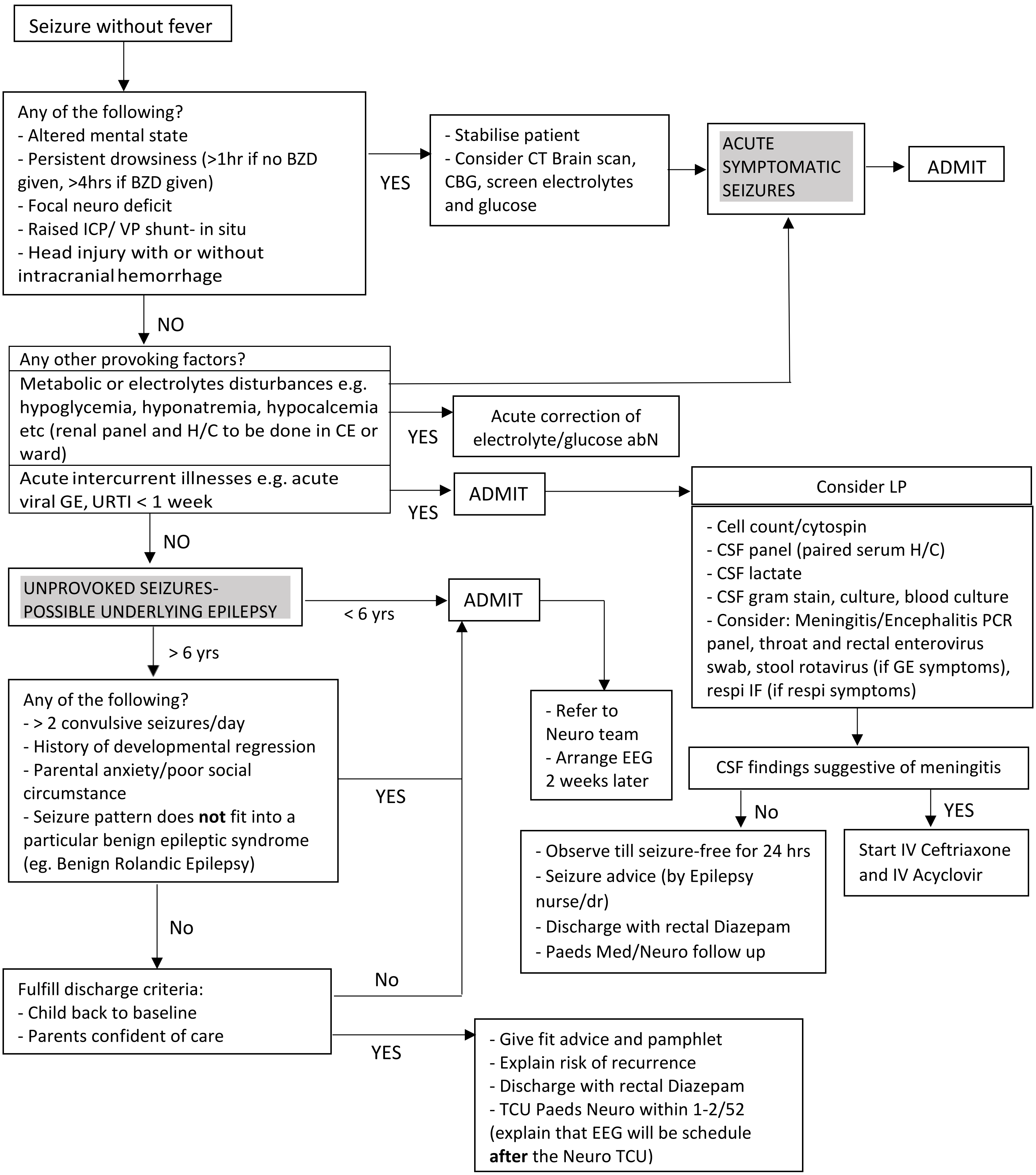
3. If febrile seizure, assess if simple or complex

4. Identify and manage the cause of the fever

5. Fever control

6. Parental reassurance and education

PAEDIATRIC AFEBRILE SEIZURES PROTOCOL



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AFEBRILE SEIZURES

Seizures are a common occurrence. Prospective population based studies show up to an 8-10% lifetime risk of one seizure, and a 1-3% chance of epilepsy. People with epilepsy have a risk of premature death that is 2-3 times higher than in the general population. Population based studies indicate that 25-30% of first afebrile seizures are provoked. If a first seizure is unprovoked, around a third to a half will recur, leading to a diagnosis of epilepsy.

Definitions

- An afebrile seizure is a seizure that occurs when the body temperature recorded at the point of seizure is less than 38 degrees Celsius. It may be classified as either acute symptomatic or unprovoked seizure.

- An acute symptomatic seizure is one that results from some immediately recognizable stimulus or cause. A diagnosis of acute symptomatic seizure should also be made in the presence of severe metabolic derangements (documented within 24h by specific biochemical or hematologic abnormalities), intracranial pathology (bleed, stroke, intracerebral neoplasm), drug or alcohol intoxication and withdrawal, or exposure to well-defined epileptogenic drugs.

- A provoked seizure is taken locally to imply an afebrile seizure associated with minor childhood infection (excluding CNS infection). We loosely use the term “provoked seizure” but this is not the term accepted internationally or synchronous with the definition.

- An unprovoked afebrile seizure is one that occurs in the absence of fever or infectious illness, and implies that the event may be an epileptic seizure in the absence of repetitive episodes required for a diagnosis of epilepsy.

- Epileptic seizures are defined as recurrent seizures due to hypersynchronous electrical discharges from the grey matter of the cerebral cortex, for which EEG may give supportive evidence to the epilepsy syndrome diagnosis.

Management

1. Control seizure:

- IV Lorazepam 0.05- 0.1mg/kg/dose over 2 mins

- IM Midazolam 0.2mg/kg/dose if there is no intravenous access

- Rectal Diazepam 0.3-0.5mg/kg/dose if there is no Intravenous access

- IV phenytoin or IV phenobarbitone 15 – 20 mg/kg/dose loading

- if seizures continue despite IV lorazepam, refer to Status Epilepticus protocol